

POTENTIAL EXPOSURE TO PFCs IN THE HMONG COMMUNITY IN THE EAST METRO

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TRiO RONALD E. McNAIR POST-BACCALAUREATE ACHIEVEMENT PROGRAM



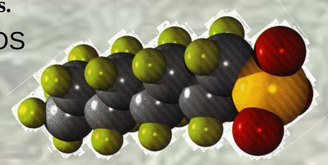
ABSTRACT

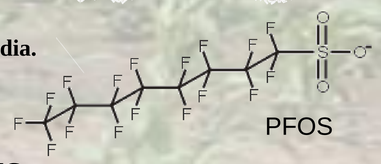
Increasing concentration levels of perfluorochemicals (PFCs) in the environment and biological material has become a growing focus of research. There is much to still be understood about the effects of these chemicals on humans and the environment. From their extremely stable chemical state, to their bio-accumulative tendency, PFCs pose serious implications for humans. The 3M-disposal site in close proximity to the Hmong farmlands poses a potential risk of exposure to these chemicals. This risk may not be realized since the farmlands are rented and not owned by the Hmong farmers. There lies a knowledge gap as to whether these lands are safe to farm, and if produce is safe for consumption. In collaboration with the community, we sampled soil and vegetation on these lands. We analyzed samples using LC/MS/MS and measured for concentration levels of PFC (detection limits: 10ppb for plants, 1.33ppb for soils). Our results did not show elevated levels of PFCs in the samples. All samples were below the detection limits. These study results should only be the beginning of deeper and larger scaled studies in this area to comprehensively determine the true status of the East Metro farmlands. Elevated PFC levels will have implications in how industries dispose of their waste. Conversely if levels are low, this should positively impact the market of the farmers as well as securing food safety for buyers.

BACKGROUND

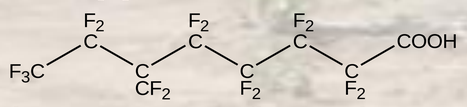
- **Perfluorocarbons (PFCs) are a class of anthropogenic chemicals that are globally distributed in environmental and biological media.**
- **They are very stable and persistent-do not biodegrade and bioaccumulative.**
- **Perfluorooctane sulfonate (PFOS) and Perfluorooctanoic acid (PFOA)-two types of PFCs, are fluorosurfactants used in manufacturing of consumer products such as stain and water repellants, food contact paper, microwave popcorn bags, fast food wrappers etc.**

PFOS





PFOA



- **Hmong farmers renting lands in the East Metro are contributors to the urban agriculture and produce for Minneapolis.**
- **These farmers are being missed in bio monitoring efforts creating challenges to their safety and product marketability.**
- **There has been continued speculation and concern that these lands have been contaminated by the nearby 3M manufacturing and disposal site.**
- **This study looks at the PFC concentration levels in the soils and produce from these farms.**

METHODS

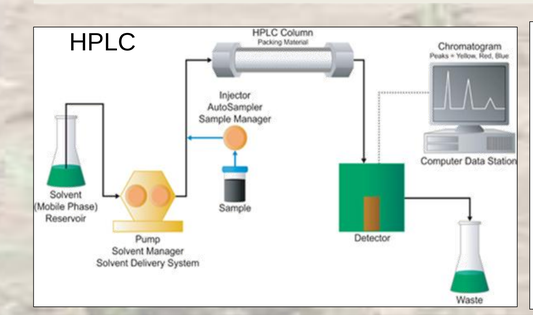
- **Prior to sampling, met with the Hmong Community Steering Committee to discuss study implications and suitable locations for sampling.**

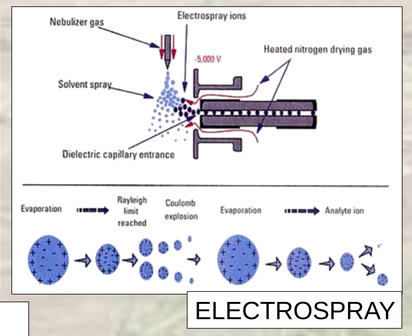


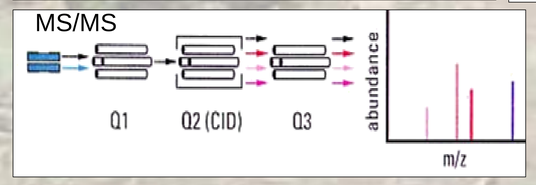


- **Plant samples specifically chosen to include staples in the Hmong diet, such as Zuab ntsuab (Mustard greens) and Dej tshuaj (Medicinal tea).**

Plants	Soil
Sampling	
▪ Sampled by hand ▪ 50ml centrifuge tubes	▪ Soil corer ▪ surface – 3ft deep. ▪ ashed glass jars.
Extraction	
▪ 2-5g of plant ▪ 20 ml of MeOH ▪ 50ml centrifuge tubes ▪ Homogenized ▪ Wrist shaker 3 hours ▪ Centrifuge 30 min. ▪ 110ug Internal Standard ($^{13}C_4$PFOS)	▪ 15g homogenized soil ▪ 30ml of MeOH ▪ 50ml centrifuged tubes ▪ Wrist shaker 3 hours ▪ Centrifuge 30 min. ▪ 110ug Internal Standard ($^{13}C_4$PFOS)
Analysis	
▪ LC/MS/MS – using relative response factor method to determine mass	

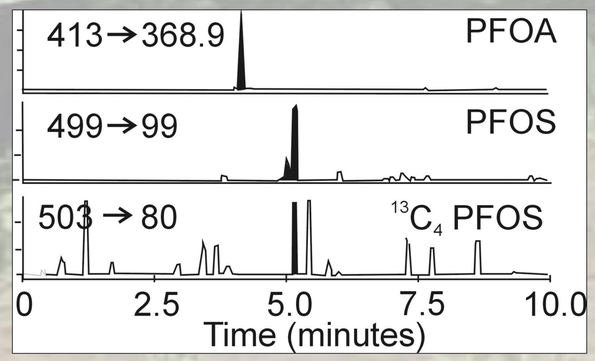






RESULTS

- **Results did not show elevated levels of PFCs within our detection limits.**
- **This does not mean that there are no PFCs in the area but that we were not able to detect them with these detection limits.**



DISCUSSION

- **Study limitations include:**
  - **-Limited number of soil and plant samples**
  - **-Single location**
  - **-Relatively high detection limit**
  - **-Presence of analytes in blanks**
- **Further study and continued sampling and testing is needed in the area.**
- **Implications of PFC free soil for this community would be a direct economic advantage, as well as securing food safety.**

REFERENCES

Jennifer G. Sepulvado†, Andrea C. Blaine†, Lakhwinder S. Hundal‡, and Christopher P. Higgins. Occurrence and Fate of Perfluorochemicals in Soil Following the Land Application of Municipal Biosolids. Environ. Sci. Technol., 2011, 45 (19), pp 8106–8112.

Washino, N., Saijo, y., Sasaki, S., Kato, S., Ban, S., Konishi, K., Ito, R., Nakata, A., Iwasaki, y., Saito, K., Nakazawa, H., Kishi, R.; Correlations between Prenatal Exposure to Perfluorinated Chemicals and Reduced Fetal Growth. Environmental Health Perspectives, Apr 2009, Vol. 117, No 4, p660-666.

Photo Credit to Suyapa Miranda, Hewlett Packard LC/MS/MS Primer and Matt Simcik.